

Merged mathematical equations of atmosphere and pollution

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September 26, 2017

Abstract

This document describes the outline of this project's plan.

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1 Equations of the atmosphere

The 3-dimensional equations of the atmosphere. Two horizontal dimensions, one vertical dimension, mass continuity, equation of state and amount of water.

2 Equations describing pollution

Three-dimensional species-continuity equation from the ScienceDirect article describing the Bulgarian pollution event.

3 Creating a synchronized system of an atmosphere containing pollution

The first two chapters describe two independent / autonomous 3D systems in different situations, having independent solutions (if they do). The first task is to merge the two models coming from different places so that they make sense in a jointed sense (still in 3D), and have common variables, common solution.

4 Dimension reduction for the jointed system

The 3D model is too complex to work with, on the other hand the phenomena can be seen as quasi-horizontal, which is why we perform dimension reduction for the merged system, the goal is to have a synchronized 2D system describing pollution and atmosphere.

5 Solutions for the jointed two-dimensional system

In this chapter we investigate solutions for the new, two-dimensional merged system.

6 Numerical simulations for the jointed two-dimensional system

Finally we perform numerical simulations.

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